

Hosna Habibi Baghi

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Professional Summary

Machine Learning Engineer specializing in NLP and production AI systems, with experience building chatbot and contact-center pipelines using modern LLM frameworks (LangChain and LlamaIndex), as well as retrieval-augmented generation. Strong background in software engineering, model evaluation, and scalable AI system development.

Education

Concordia University

M.Sc., Computer Science

Montreal, Canada

Thesis: *Verification of Temporal Properties in Timed Event Systems (Software Formal Methods)*

University of Guilan

B.Sc., Computer Engineering

Guilan, Iran

Thesis: *Blood Glucose Prediction Using Mixture of Experts*

Work Experience

Software Formal Methods Researcher

Montreal, Canada

ER Research Center of Software and AI

Dec 2024 – April 2026

- Conducted research on verification of temporal properties in timed event systems, with a focus on formal methods, event traces, and system correctness.
- Developed a simulation framework to transform UML state machines into Prolog-based representations, enabling automated reasoning and behavioral verification.
- Designed and implemented a Python–Prolog evaluation engine to query Linear Temporal Logic (LTL) properties over event traces.
- Built analysis workflows to validate dynamic and event-driven system behavior against temporal specifications.

Machine Learning Developer

Tehran

Prism

Jul 2024 – Dec 2024

- Designed and deployed NLP components for an intelligent contact-center platform, using transformer-based models and LLMs to enhance customer–agent interactions.
- Built scalable conversation analysis pipelines for intent detection, semantic search, and information extraction across large-scale support data.
- Developed and optimized LLM-based solutions using fine-tuning and retrieval-augmented generation (RAG) to improve response quality and contextual understanding.
- Established evaluation frameworks combining automated metrics and human feedback to assess conversational performance, reliability, and relevance.
- Integrated ML services into production through scalable APIs and data pipelines in collaboration with product and backend teams.
- Drove iterative experimentation and model optimization using real-world data to improve system performance and user experience.

Machine Learning Engineer

Tehran

FlowChat

Sep 2023 – Jul 2024

- Developed NLP models for production chatbot systems using BERT and sequence models (RNNs, LSTMs).
- Applied prompt engineering and experimented with LLMs for response generation and topic modeling.
- Designed end-to-end ML pipelines for training, evaluation, and deployment of conversational models.
- Integrated AI components into backend systems, ensuring scalability and software reliability through cross-functional collaboration.
- Improved chatbot performance through iterative refinement using real-world interaction data and user feedback.

AI Researcher

Tehran

University of Guilan

Sep 2022 – Sep 2023

- Conducted research in NLP and conversational AI, focusing on text understanding and chatbot systems using deep learning methods.
- Developed and evaluated neural models, including RNNs, LSTMs, and transformer-based architectures (e.g., BERT), for text classification, question answering, and response generation.
- Explored large language models (LLMs) through prompt engineering and few-shot learning for multi-turn dialogue and topic analysis.
- Built end-to-end experimental pipelines for data preprocessing, model training, and evaluation, enabling systematic comparison of NLP approaches.

Publications

Hosna Habibi Baghi, Ali Jannatpour, Constantinos Constantinides. "Verification of Linear Temporal Properties over Timed Events." *International Conference on Software Engineering and Development (ICSED 2025)*, Springer LNCS, 2025.

Skills

Programming: Python, Java, C++

Machine Learning & NLP: PyTorch, TensorFlow, Transformers, Large Language Models (LLMs), Fine-tuning (LoRA, PEFT), Prompt Engineering, Retrieval-Augmented Generation (RAG), Text Embeddings, Semantic Search

LLM Frameworks & Tooling: LangChain, LangGraph, LlamaIndex, Vector Databases (FAISS, Pinecone), Agent-based Systems

NLP Techniques: Text Classification, Named Entity Recognition (NER), Information Retrieval, Sequence Modeling

Data & Analysis: NumPy, Pandas, Data Preprocessing, Statistical Analysis, Data Visualization (Matplotlib, Seaborn)

MLOps & Deployment: Model Deployment, API Development (REST), Docker, CI/CD, Experiment Tracking (MLflow, Weights & Biases), Model Monitoring, Pipeline Development

LLM Evaluation & Optimization: Prompt Evaluation, Offline/Online Evaluation, A/B Testing, Latency & Cost Optimization, Hallucination Mitigation

Software Engineering: System Design, Testing & Validation, Performance Optimization, Version Control (Git), Linux/Unix

Web Technologies: React, JavaScript, Bootstrap